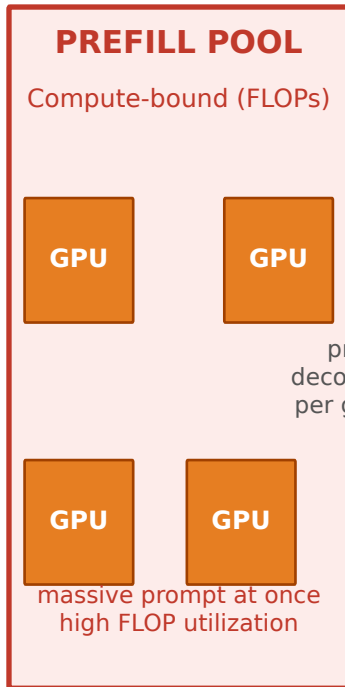


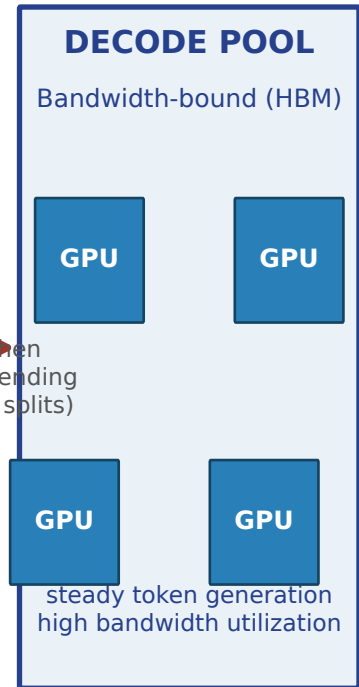
P/D Disaggregation: the two pools want opposite resources

input prompt →



**KV CACHE
TRANSFER (initial)**

prefill writes the prompt KV, then
decode READS it AND keeps appending
per generated token (ownership splits)



← tokens out to client

Why split? prefill ~1.19 PFLOPS vs 0.989 peak (FLOP-starved);
decode 5.6 TB/s vs 3.35 TB/s (bandwidth-starved). One pool forces a compromise. [ILLUSTRATIVE][DERIVED]